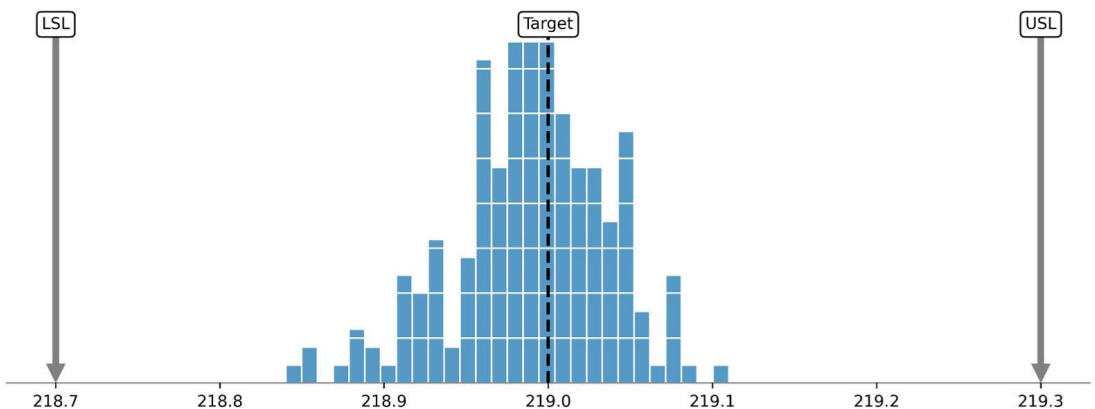


BROKEN QUALITY PRESS PRESENTS

What the spec?

The practical differences between process limits and specification limits

An essay by Jim Lehner



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What the Spec?

*The practical differences between process limits and
specification limits*

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Second edition

Contents

What the spec?	1
The practical differences	1
The impact of ignorance	2
Spec limits do not belong on process behavior charts	6
How to visualize process data and spec limits	10
Summary	14
Additional figures	16
Notes	17

What the spec?

The practical differences

The characteristic features of process behavior charts are the process limits. These calculated limits define how large or small a value must be before it represents a departure from the historic mean. When all the values on a process behavior chart fall within the process limits, the process is characterized as predictable. This is evidence that the process is under the influence of only common causes of routine variation. When one or more values on a process behavior chart fall outside the process limits, the process is characterized as unpredictable. This is evidence that the process is under the influence of both common causes of routine variation and assignable causes of exceptional variation. Regardless of the characterization and regardless of the types of variation that are present, process limits act as the *Voice of the Process*. They tell you what the process will do in the future based on what it has done in the past.

In contrast to process limits are specification limits or “spec” limits. Unlike process limits, which are values calculated from process data, spec limits are boundaries that determine when a product, process, or service is acceptable and when it is not. Spec limits are determined by the customer and, as such, are often referred to as the *Voice of the Customer*. Whether the customer is an individual, a department, or industry, spec limits tell you when it is time to take action based on what a customer will accept. What spec limits do not tell you, however, is what actions to take. They are incapable of characterizing a process

and they are incapable of articulating the types of variation that influence process behavior. They are, for better or for worse, goal posts composed of hopes and dreams.

Spec limits:	Process limits:
Voice of the Customer	Voice of the Process
Based on hopes & dreams	Based on data
Incapable of prediction	Facilitate prediction
Can't direct time & attention	Direct time & attention

Fig 1: The differences between spec limits and process limits.

The impact of ignorance

Those that are ignorant of the differences between process limits and spec limits reliably conflate the two. While these individuals can be excused for their lack of knowledge, they cannot be excused for the havoc their lack of knowledge wreaks. Similar to the consequences that result from making either one of the two types of mistakes¹, confusing process limits with spec limits or spec limits with process limits results in misguided demands that cause misguided actions.

Figure 2 shows the process behavior chart of weights for 26 carbon fiber kit aircraft elevators—elevators are the flight control surface that controls an aircraft's pitch. These parts were produced sequentially by the same process. If the process limits in Figure 2, the *Voice of the Process*, are confused for spec limits, the *Voice of the Customer*, what actions result?

When we interpret the process limits in Figure 2 as spec limits, parts with weights that fall outside the limits will be rejected. That is to say, parts with weights outside the limits will be deemed unacceptable for use by the customer. Given this criteria, only part E19 is deemed unacceptable in Figure

¹ Recall that Mistake 1 confuses assignable causes of variation for common causes of variation and Mistake 2 confuses common causes of variation for assignable causes of variation.

WHAT THE SPEC?

2. Subsequently, the fate of part E19 sits at the crossroads of scrap or rework. If the part is designated for scrap, the full cost of producing the part will be incurred as a loss. If the part is designated for rework, the effective cost of producing the part will increase but incurring the full cost of producing the part as a loss will be avoided.

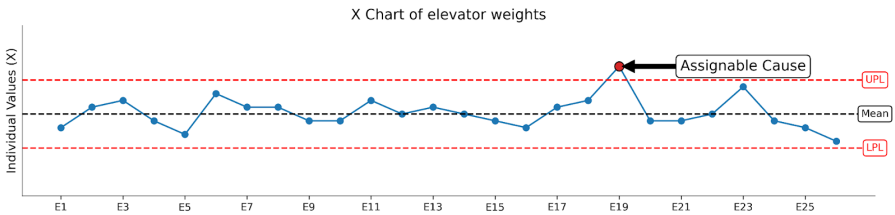


Fig 2: X Chart of kit aircraft elevator weights with one value greater than the upper process limit.

The prior actions would be well and good if the limits displayed in Figure 2 were indeed spec limits. But remember, Figure 2 is a process behavior chart. Specifically, it is the X Chart portion of an XmR Chart. The horizontal lines displayed on this chart are not spec limits. They are process limits. When interpreting a process behavior chart, like the one shown in Figure 2, if one or more values fall outside the process limits the process is characterized as unpredictable. Inherent in this characterization is the knowledge that the process is influenced by assignable causes of exceptional variation.

In Figure 2, Part E19 has a weight greater than the Upper Process Limit. This is evidence that assignable causes influence process behavior. The proper path forward should subsequently work to identify, understand, and eliminate the source of this exceptional variation, not ruminate over whether to scrap or rework the part.

When we confuse the process limits in Figure 2 for spec limits we make the first type of mistake—Mistake 1. Mistake 1 occurs when we confuse assignable causes of exceptional variation for common causes of routine variation. This is the same type of mistake that occurs when we rely on spec limits alone to

understand process behavior. Figure 3 acts as an example.

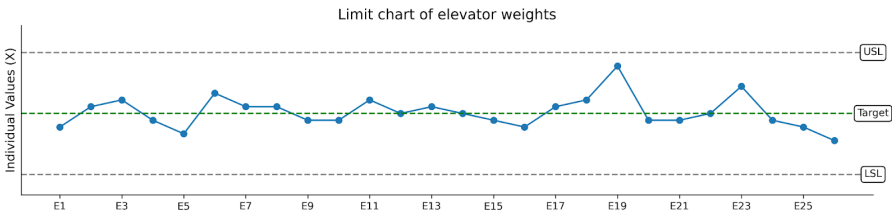


Fig 3: Limit chart of elevator weights shows that all of the values fall within the specification limits.

In Figure 3, the same 26 parts from Figure 2 are displayed in the context of the associated spec limits. From this perspective, the *Voice of the Customer* deems all of the parts acceptable because all of the weights fall inside the spec limits. However, the process behavior chart in Figure 2 characterized this process as unpredictable. Such a characterization indicates that future process behavior cannot be predicted within limits. While today the process is satisfying customer requirements, the unpredictable characterization from the process behavior chart acts as a warning that this may not last. There should be no expectation that parts will continue to be produced inside the spec limits. As such, it is better to identify, understand, and eliminate the influence of the assignable cause before additional costs are incurred. To not do so is to wager with the economic viability of the process and, perhaps more importantly, risk damaging the relationship with the customer.

Where Figures 2 and 3 articulate the consequences of confusing process limits for spec limits, Figure 4 provides an example of confusing spec limits for process limits. In Figure 4, the weights of 20 carbon fiber kit aircraft ailerons—ailerons are the flight control surface that controls an aircraft's roll—are shown in the context of the associated spec limits. If these spec limits, the *Voice of the Customer*, are confused for process limits, the *Voice of the Process*, what actions result?

In the lexicon of process behavior charts, a process with one or more values

WHAT THE SPEC?

that fall outside the process limits is characterized as unpredictable. Inherent in this characterization is the knowledge that the process displays exceptional variation due to assignable causes. Part A6, which has a weight less than the Lower Spec Limit, and part A18, which has a weight greater than the Upper Spec Limit, when interpreted under the guise of a process behavior chart acts as evidence that this process is influenced by exceptional variation. They act as evidence of assignable causes. Subsequently, the time and attention of technical resources will work to understand the underlying cause. Individuals and the team will work to eliminate the source of this exceptional variation.

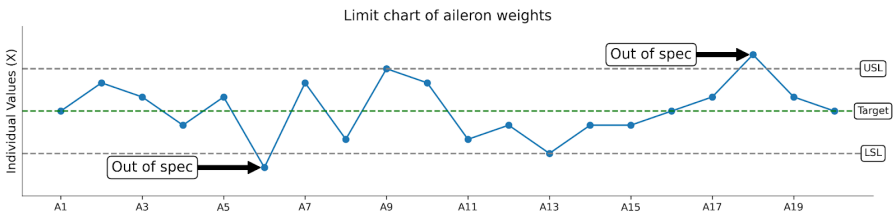


Fig 4: The limit chart for aileron weights shows two parts, A6 and A18, are out of spec.

However, Figure 4 is not a process behavior chart and the horizontal lines are not process limits. The horizontal lines shown in Figure 4 are spec limits. To confuse these spec limits for process limits is to make the second type of mistake—Mistake 2. It is to confuse common causes of routine variation for assignable causes of exceptional variation. As such, any edicts that demand explanations for what are perceived as “assignable causes” are a waste of time. As shown in the process behavior chart of the same ailerons in Figure 5, the process in Figure 4 is characterized as predictable. It displays only routine variation and thus is influenced by only common causes. In order to align the *Voice of the Process* with the *Voice of the Customer* fundamental changes to the process are required. Those fundamental changes will manifest in one of the six improvement domains from which common causes of routine variation originate: Machine, Material, Method, Measurement, Environment,

or Personnel.

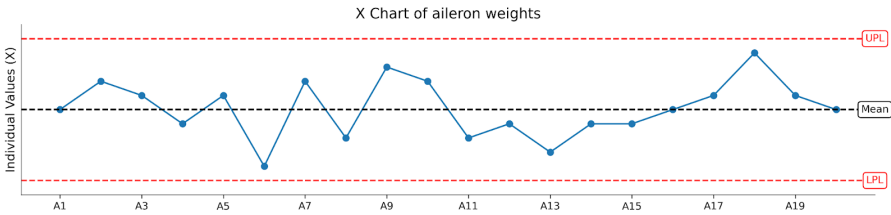


Fig 5: X Chart of kit aircraft aileron weights with all values inside the process limits.

Regardless of the specific type of confusion, whether it is confusing process limits for spec limits or spec limits for process limits, these types of mistakes have real consequences. These consequences can be avoided with clear and concise operational definitions.

Spec limits are the *Voice of the Customer*. They are the boundaries between what a customer will accept and what a customer will reject. Process limits, on the other hand, are the *Voice of the Process*. They tell you what a process will do in the future based on what it has done in the past. While, in an ideal world, spec limits and process limits would always agree, they are under no obligation to do so. It is up to you, through methodical, thoughtful, and continuous improvement efforts to ensure that processes reliably and repeatedly produce conforming product and customers get what they are asking for. It is up to you, through the use of process behavior charts, that every effort is made at every turn to eliminate the influence of variation.

Spec limits do not belong on process behavior charts

The confusion that results from the lack of knowledge regarding differences between spec limits and process limits is amplified due to lack of standards in terms of their visual display. Rather than dictate when, where, and how to visually display process data in the context of spec limits, industry has left the

task open ended. This has birthed a vast array of visualization options that only sometimes achieve their goals. On the off chance that data visualization efforts do achieve their goals, they do so at an audience's expense in the form of comprehension. Rather than impart context subliminally through preattentive processing, most visualizations that incorporate spec limits require interpretation. They require the overt recognition by an audience that the limits being displayed are the *Voice of the Customer* not the *Voice of the Process*. This slows the time to insight for an audience and undermines an author's efforts to turn the data into insights and those insights into actions. It injects friction into the process of understanding variation where it should be well lubricated.

The most common but egregious visual mistake regarding spec limits and process limits is shown in Figure 6. Here, in addition to the process limits and mean, the process behavior chart is thought to be enhanced by the inclusion of spec limits. Shown as green dotted lines, the Upper Spec Limit (USL) and Lower Spec Limit (LSL) bound the range of values that are acceptable to the customer. Anything inside these bounds is deemed acceptable while anything outside these bounds is deemed unacceptable. While ostensibly this approach seems reasonable, its practice should be avoided for a variety of reasons with varying impact.

The first and perhaps most flagrant problem that arises from the inclusion of spec limits on a process behavior chart is how spec limits distract an audience from the purpose of using a process behavior chart in the first place. Recall that the power and utility of process behavior charts rests in their ability to put data in context. This context is born from the relationship between the values in a data set and the associated process limits. When all the values on a process behavior chart fall inside the process limits the process is characterized as predictable. When one or more values fall outside the process limits on a process behavior chart the process is characterized as unpredictable. In either case, characterization establishes a path forward. It articulates the kinds of variation that influence process behavior and, in doing so, determines where the time and attention of technical resources should be directed. This is the purpose of a process behavior chart. It is to bring focus to the *Voice of the*

WHAT THE SPEC?

Process. It is to characterize what a process has done in the past so we can anticipate what it will do in the future.

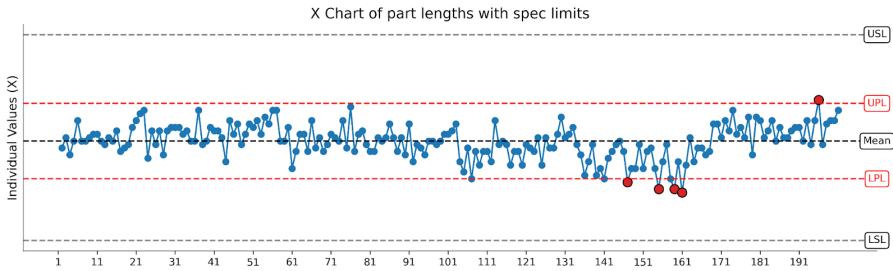


Fig 6: X Chart of part lengths with the associated spec limits. NEVER put spec limits on a process behavior chart.

However, as a purveyor of the visual display of quantitative information, it is up to you, the author, not the audience to direct attention. It is up to you, through pragmatic design choices, to highlight what is important. Choosing to use a process behavior chart to visually display process data, in and of itself, is a choice to direct an audience's attention to variation. It is a choice to focus on the underlying nature of a process as either predictable or unpredictable. It is a choice to highlight the *Voice of the Process*. Including spec limits on a process behavior chart undermines this choice. It pulls attention away from the *Voice of the Process* and stretches it to include the *Voice of the Customer*. Rather than conflate the two domains in a single figure, you and your audience benefit from exploring these two voices separately.

As an example, the same data from Figure 6 is presented again in Figure 7. This time, however, Figure 7 does not include the spec limits. Given this alternative visual display of the process data, how does your interpretation of the process behavior chart change? What kind of questions do you ask with regard to Figure 7 that you would not have asked with regard to Figure 6?

WHAT THE SPEC?

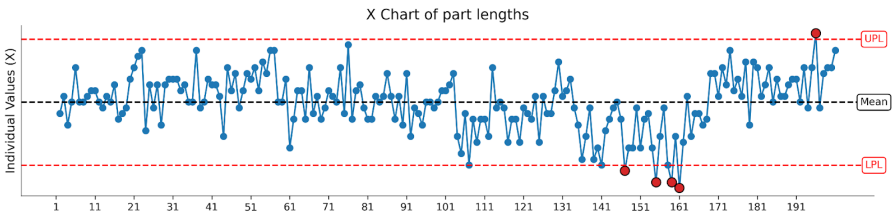


Fig 7: X Chart of part lengths WITHOUT spec limits brings focus to the Voice of the Process.

Unlike Figure 6 which divides the audience's attention between the *Voice of the Process* and the *Voice of the Customer*, the process behavior chart in Figure 7 focuses on the *Voice of the Process*. It makes clear to the audience that, regardless of the spec limits, past process behavior is characterized as unpredictable. Inherent to this insight is the knowledge that both common causes of routine variation and assignable causes of exceptional variation are present. As such, the time and attention of technical resources must work to identify, understand, and eliminate the assignable causes.

Adding spec limits to the process behavior chart, as is the case with Figure 6, reduces these dynamic insights to a binary discussion of good or bad. It implies that because all of the parts that are produced by the process fall within the spec limits no work is required. It indicates that, regardless of the presence of assignable causes, no improvement is necessary. But improvement is necessary. Assignable causes of exceptional variation are acting on this process and, as such, the process is characterized as unpredictable. Regardless of whether or not these assignable causes are affecting the customer today, their presence indicates there is potential for them to impact the customer in the future. To not take action now, even in the face of this evidence, is to wager with the future economic viability of the process and the relationship with the customer.

The second problem that emerges when spec limits are included on a process behavior chart is the product of business and industry. The lack of exposure, knowledge, and use of process behavior charts, in conjunction with the lack of visual standards for when, where, and how to visually display

spec limits, leaves many audiences ill prepared to interpret process behavior charts without guidance. While this is not an insurmountable task, it does impart some friction. This friction manifests in the form of explanations that would not be necessary if business and industry were more familiar with the application, use, and interpretation of process behavior charts.

Rather than compound efforts to understand and explain the *Voice of the Process* and the *Voice of the Customer*, time and attention is better spent by exploring the concepts separately. This gives individuals unfamiliar with process behavior charts the opportunity to digest their newfound knowledge. It gives them time to sit with the implications of this analytical approach and comprehend the value of understanding variation.

How to visualize process data and spec limits

The prior discussion regarding the confusion that results from including spec limits on a process behavior chart naturally turns to inquiry about how to display process data and spec limits together. The answer to this question varies depending on the context but is limited to two forms: the histogram and the time series. While both of these visualizations achieve the same goal of relating the process data to the requisite spec limits, they do so in markedly different ways. Where a histogram collapses the time base element of a data set, time series expand it. Where histograms reveal the shape of the data's distribution in the form of shape, center, and spread, time series reveal the interesting information that is contained within the time-order sequence of the data.

WHAT THE SPEC?

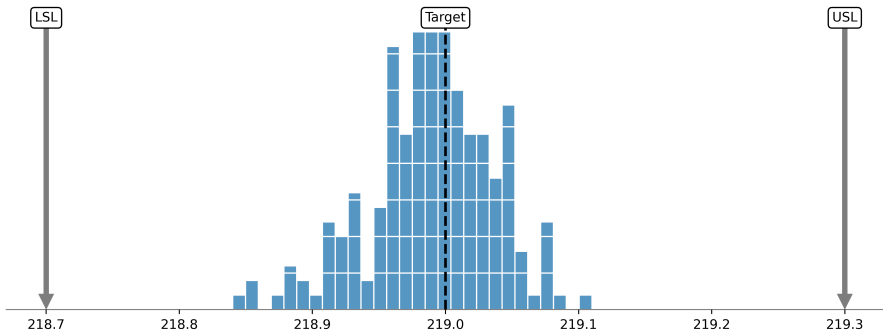


Fig 8: The histogram of part lengths displayed in the context of the spec limits and the process target.

The process data from Figure 7 is shown in the form of a histogram that includes spec limits in Figure 8. Often referred to as a *capability histogram*, the visual format of Figure 8 puts the process data in the context of the *Voice of the Customer* rather than the *Voice of the Process*. Bound on the left by the Lower Spec Limit (LSL) and the right by the Upper Spec Limit (USL) the histogram in Figure 8 makes clear that this process is producing product that the customer will accept. The distance between the spec limits, called the tolerance, is bifurcated by the target, shown as a green dotted line. Subsequent interpretation of Figure 8 is simple and need not be further complicated. All values fall within the spec limits. Nothing more and nothing less. While discussions about the shape of the distribution are likely to ensue, such discourse provides no insight into the behavior of the underlying process. Any statements that are made about process behavior, based on this figure, are nothing more than conjecture. This is because unlike process behavior charts, histograms are incapable of discriminating between the two types of variation. They are incapable of characterizing a process as either predictable or unpredictable.

The alternative form for displaying process data in the context of spec limits is shown in Figure 9. Here, the data is displayed as a time series in the form of a limit plot. Rather than combine the values in the data set to reveal the shape of the distribution, as was the case for the histogram in

Figure 8, the time series in Figure 9 forces each value to stand on its own. Like Figure 25.8, the Upper Spec Limit and Lower Spec Limit are shown as gray dotted lines while the target is shown as a green dotted line. The resulting interpretation of Figure 9, as was the case with the histogram in Figure 8, is simple. The limit plot in Figure 9 shows that all values fall within the spec limits. Nothing more and nothing less. While discourse asking why some values are above the target and some values are below the target is likely, such discussion is a waste of time. Unlike process behavior charts, but like histograms, time series are incapable of discriminating between the two types of variation. They are incapable of characterizing a process as either predictable or unpredictable. Any statements that are made regarding the behavior of the underlying process, based on the time series in Figure 9 alone, are based on speculation and should not be trusted.

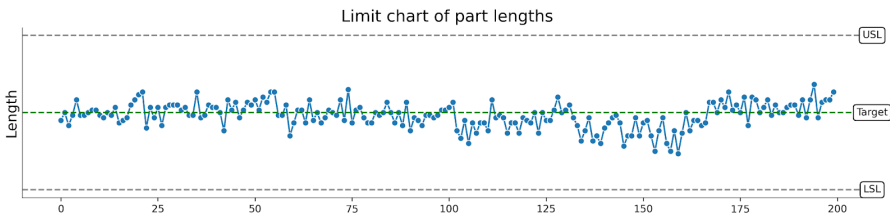


Fig 9: The limit chart of part lengths.

Displaying data in the form of the histogram in Figure 8 or the time series in Figure 9 brings discrete focus to the *Voice of the Customer*. These visual displays establish that the process is producing a product that is acceptable. From this perspective, no immediate action is needed since there is no loss of product or profit. However, just because no immediate action is needed does not mean no actions should be taken.

The process behavior chart in Figure 7 revealed several values fell outside the process limits. This characterizes the process as unpredictable and indicates that both common causes of routine variation and assignable causes of exceptional variation are present. Regardless of whether or not this impacts

the customer today, an opportunity for improvement is present. To not take action on this opportunity is to ignore the *Voice of the Process*. It is to openly disregard a signal from the process that something out of the ordinary has occurred. This closes the door on a chance to learn something about the process. Such knowledge could prove invaluable in mitigating the future impact of assignable causes on production and the customer.

Even though histograms and time series are both adept at displaying process data in the context of spec limits, an unwritten convention has been presumed by industry. This convention prioritizes the use of histograms over time series for the visual display of process data in the context of spec limits. The practical and psychological utility of this approach is one and the same. By religiously displaying spec limits only in the context of histograms, a visual presupposition is established with the audience. This presupposition primes an audience to expect that, when applicable, histograms of process data will also display spec limits. Repeated exposure to this format is a signal to the audience that histograms with vertical lines can be assumed to be the *Voice of the Customer* while time series with horizontal lines (i.e. process behavior charts) can be assumed to be the *Voice of the Process*.

This convention also holds utility for you as an analyst, author, and communicator. By standardizing your method of spec limit visualization, the task of visualizing and communicating about the *Voice of the Customer* is simplified. Rather than reinvent your approach with each and every effort, a toolkit composed of predefined methods is established. While this is likely to require modifications depending on the process and circumstance, having a foundation upon which you can build will streamline your efforts. It ensures iteration, learning, and continuous improvement, as articulated in the PDSA cycle, is not only reflected in improvement efforts but your work as a communicator and professional.

Multi-chart process plots

In instances where it is advantageous or necessary to present both the *Voice of the Process* and the *Voice of the Customer* at the same time, a multi-chart process plot like the one shown in Figure 10 can prove useful. Here, the process data is displayed in the context of both a process behavior chart and histogram with spec limits (capability histogram). This visual display allows the audience to review and interpret both charts independently and via comparison. For the process in Figure 10, the multi-chart process plot highlights that, although all the values fall within the spec limits on the histogram, the process is characterized as unpredictable by the process behavior chart. Thus, in order to avoid losses in the future, time and attention should be dedicated to understanding the assignable causes of exceptional variation influencing this process.

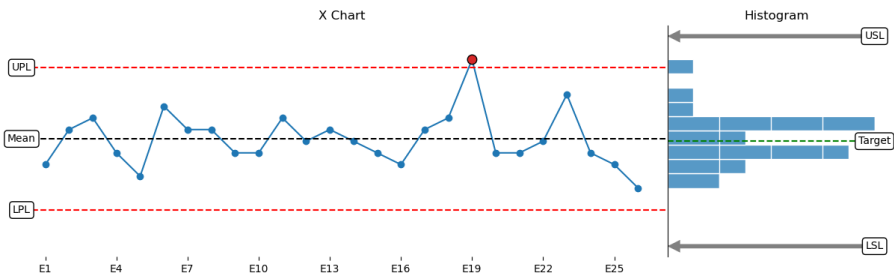


Fig 10: Multi-chart process plot of elevator weights simultaneously displays the X Chart and capability histogram.

Summary

The differences between spec limits and process limits are not widely understood or appreciated. This is more a product of ignorance than it is malice. Where spec limits are the *Voice of the Customer* process limits are the *Voice of*

the Process. This distinction is made more clear by understanding the origin of each set of limits.

As the *Voice of the Customer*, spec limits are established by the customer. They are a set of limits that tell you what values are acceptable and what values are unacceptable. This means, for better or for worse, spec limits are hopes and dreams. They are not calculated from process data and, subsequently, do not reflect process behavior. They are based on what a customer wants to see. They tell you when it is time to take action based on what a customer will accept, but they do not tell you what actions to take. To know what actions to take you must turn to the *Voice of the Process*.

Process limits are the *Voice of the Process*. They are a set of calculated limits that determine how large or small a single value must be before it represents a departure from the historic mean. Unlike spec limits, which are based on hopes and dreams, process limits are based on data. As such, future process behavior can be predicted using process limits and specific actions to alter process behavior can be taken.

Confusing spec limits for process limits or process limits for spec limits has consequences akin to making either one of the two types of mistakes. Such confusion directs the time and attention of technical resources to where they are not needed. This undermines improvement efforts and relegates their success to luck rather than analysis.

Although it is tempting to display spec limits on a process behavior chart, such a display undermines the purpose and aim of the process behavior chart. The choice to visualize process data using a process behavior chart is a choice to draw an audience's attention to the *Voice of the Process*. It is a choice to highlight process variation. Placing spec limits on a process behavior chart mutes this choice. It distracts an audience from the underlying analytical effort that aims to focus on the *Voice of the Process* and understand variation. It is for this reason that spec limits should never be placed on a process behavior chart. Instead, when it is necessary to display process data in the context of spec limits, a histogram can be used. However, even when a histogram is used, a process behavior chart of the data should also be presented. This ensures that the *Voice of the Process* and the *Voice of the Customer* both have a seat at

the table.

Additional figures

For increased clarity in comparison of Figures 2 and 3, as well as increased clarity for comparison of Figures 4 and 5, Figures 11 and 12 are included as reference. Unlike Figures 2 and 3 and Figures 4 and 5 which plotted the two sets of data independently, the two halves of Figures 11 and 12 plot the same data concurrently. What differs is the context. The left side of each figure below displays the data in the context of the process limits using an X Chart while the right side of each figure below displays the data in the context of the specification limits.

The shared y-axis between the X Charts and limit plots for Figures 11 and 12 allows for direct visual comparison. Such comparison makes clear that, in Figure 11, the process limits (shown in red) are tighter than the spec limits (shown in gray). Conversely, Figure 12 makes clear that process limits (again shown in red) are wider than the spec limits (again shown in gray).

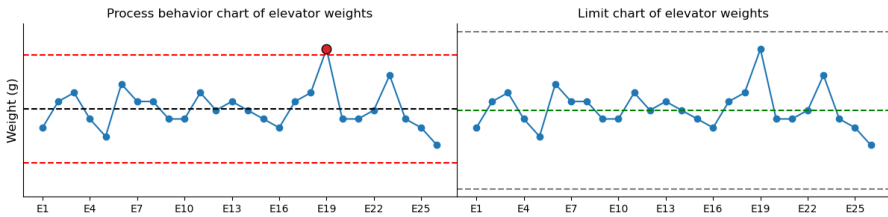


Fig 11: Elevator weights in the context of the process limits (left) and spec limits (right).

WHAT THE SPEC?

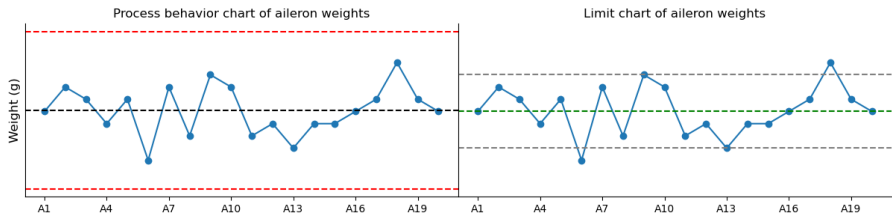


Fig 12: Aileron weights in the context of process limits (left) and spec limits (right).

Notes

- Specification limits, or spec limits, are the *Voice of the Customer*
- Spec limits establish what values are acceptable and what values are unacceptable for a customer
- Process limits are the *Voice of the Process*
- Process limits establish how large or small a single value must be before it represents a departure from the historic mean
- While spec limits tell you when it is time to take action based on what a customer will accept they do not tell you what actions to take
- Process limits tell you when it is economically viable to take action on a process and what actions to take
- Confusing spec limits and process limits can result in taking the wrong actions at the wrong time
- Spec limits do not belong on process behavior charts
- When necessary to display process data in the context of spec limits a histogram should be used
- Adding spec limits to a process behavior chart undermines the purpose of using a process behavior chart in the first place

